

Problem 1

// Task 1 : Create a structure named Student with two members : name and age. Create one variable of the structure and display its values.

```
#include <iostream>
```

```
using namespace std;
```

```
struct General
```

```
{
```

```
    string name{};
```

```
    int age{};
```

```
} myStruct;
```

```
int main()
```

```
{
```

```
    myStruct.age = 12;
```

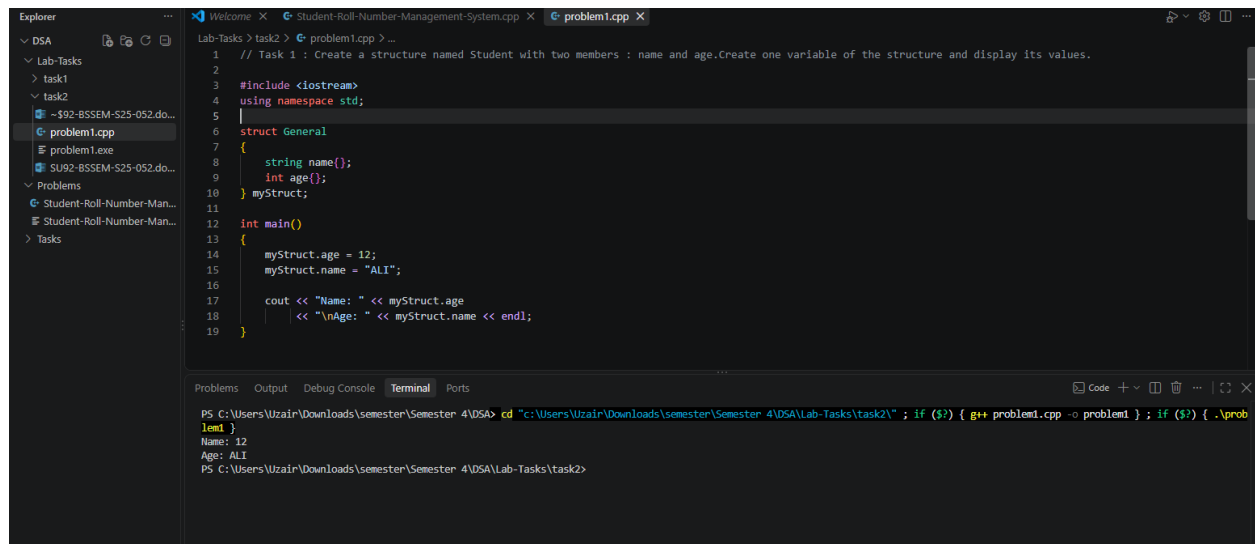
```
    myStruct.name = "ALI";
```

```
    cout << "Name: " << myStruct.age
```

```
        << "\nAge: " << myStruct.name << endl;
```

```
    return 0;
```

```
}
```



Problem 2

// Task 2 : Create one node dynamically, store a value in data, and set its next pointer to NULL.

```
#include <iostream>
```

```
using namespace std;
```

```
struct Node
```

```
{
```

```
    int data{};
```

```
    Node *next;
```

```
};
```

```
int main()
```

```
{
```

```
    Node *firstNode = new Node;
```

```
    firstNode->data = 100;
```

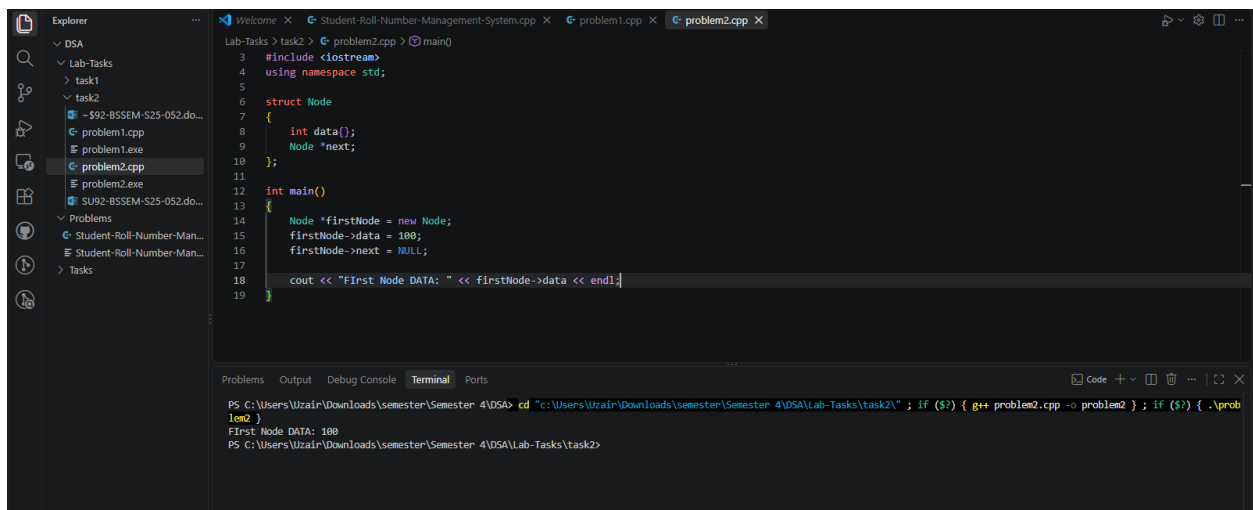
```
    firstNode->next = NULL;
```

```
cout << "First Node DATA: " << firstNode->data << endl;
```

```
delete firstNode;
```

```
return 0;
```

```
}
```



Problem 3

// Task 3 : Create three nodes containing 10, 20, and 30. Connect them in sequence and make the last node point to NULL.

```
#include <iostream>
```

```
using namespace std;
```

```
struct Node
```

```
{
```

```
    int data{};
```

```
    Node *next;
```

```
};
```

```
int main()
```

```
{
```

```
    Node *firstNode = new Node;
```

```
    Node *secondNode = new Node;
```

```
    Node *thirdNode = new Node;
```

```
    firstNode->data = 10;
```

```
    secondNode->data = 20;
```

```
    thirdNode->data = 30;
```

```
    firstNode->next = secondNode;
```

```
    secondNode->next = thirdNode;
```

```
    thirdNode->next = NULL;
```

```
    cout << "First Node DATA: " << firstNode->data << endl;
```

```
    cout << "Second Node DATA: " << secondNode->data << endl;
```

```
    cout << "Third Node DATA: " << thirdNode->data << endl;
```

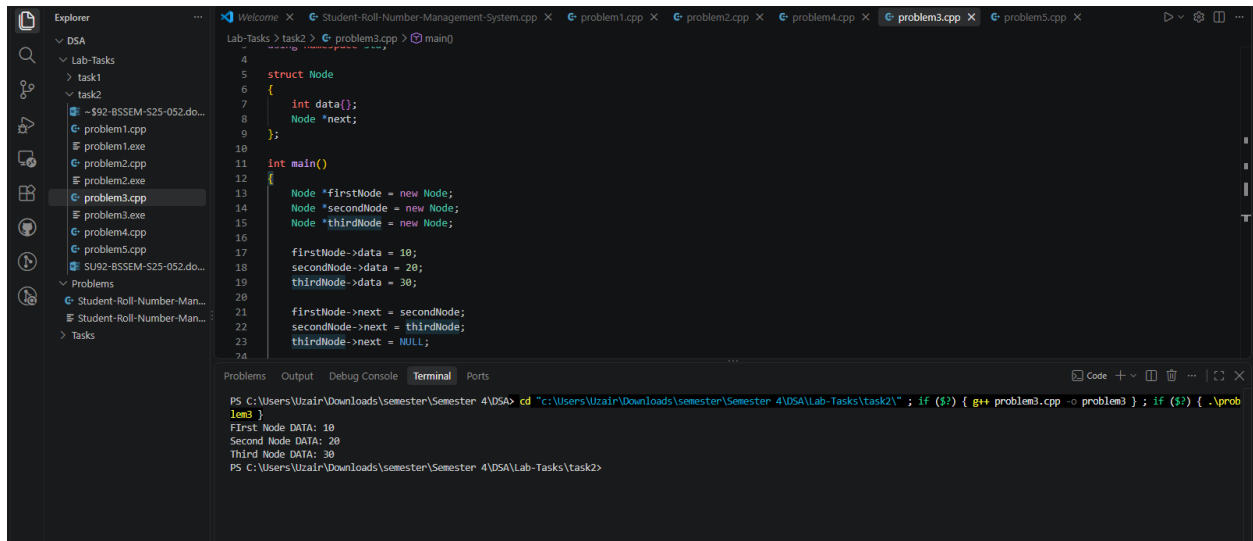
```
    delete firstNode;
```

```
    delete secondNode;
```

```
    delete thirdNode;
```

```
    return 0;
```

```
}
```



Problem 4

// Task 4 : Create three connected nodes and use a head pointer to point to the first node. Display the value stored in the first node.

```
#include <iostream>
```

```
using namespace std;
```

```
struct Node
```

```
{
```

```
    int data{};
```

```
    Node *next;
```

```
};
```

```
int main()
```

```
{
```

```
    Node *firstNode = new Node;
```

```
    Node *secondNode = new Node;
```

```
Node *thirdNode = new Node;
```

```
Node *Head = firstNode;
```

```
firstNode->data = 10;
```

```
secondNode->data = 20;
```

```
thirdNode->data = 30;
```

```
firstNode->next = secondNode;
```

```
secondNode->next = thirdNode;
```

```
thirdNode->next = NULL;
```

```
cout << "First Node DATA: " << Head->data << endl;
```

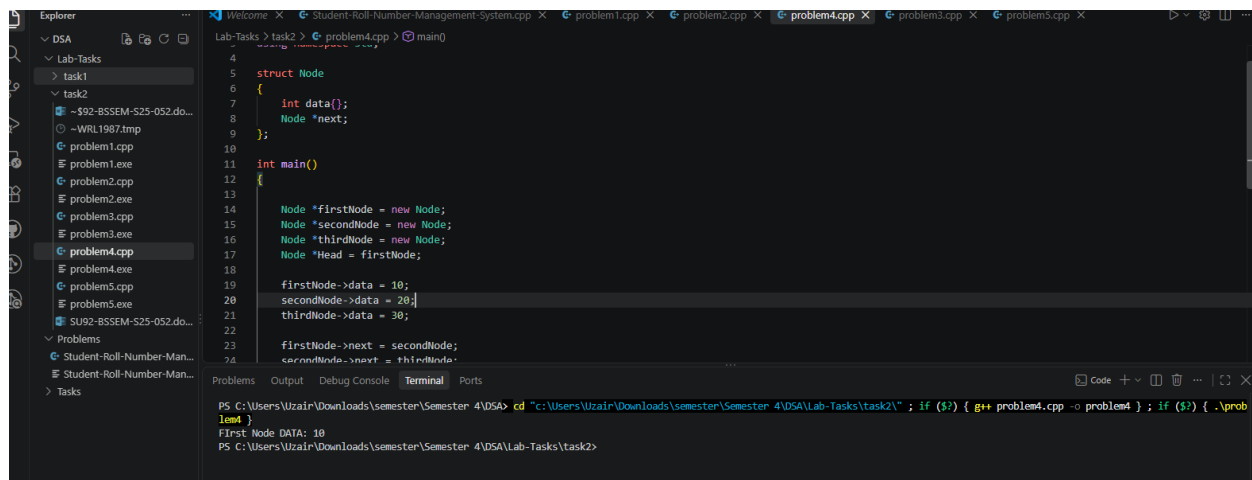
```
delete firstNode;
```

```
delete secondNode;
```

```
delete thirdNode;
```

```
return 0;
```

```
}
```



Problem 5

// Task 5 : Create two nodes representing customers 101 and 102. Connect the first customer to the second using the next pointer.

```
#include <iostream>
```

```
using namespace std;
```

```
struct Node
```

```
{
```

```
    int data{};
```

```
    Node *next;
```

```
};
```

```
int main()
```

```
{
```

```
    Node *customer101 = new Node;
```

```
    Node *customer102 = new Node;
```

```
    customer101->data = 101;
```

```
    customer102->data = 102;
```

```
customer101->next = customer102;
```

```
customer102->next = NULL;
```

```
cout << "Customer: " << customer101->data << endl;
```

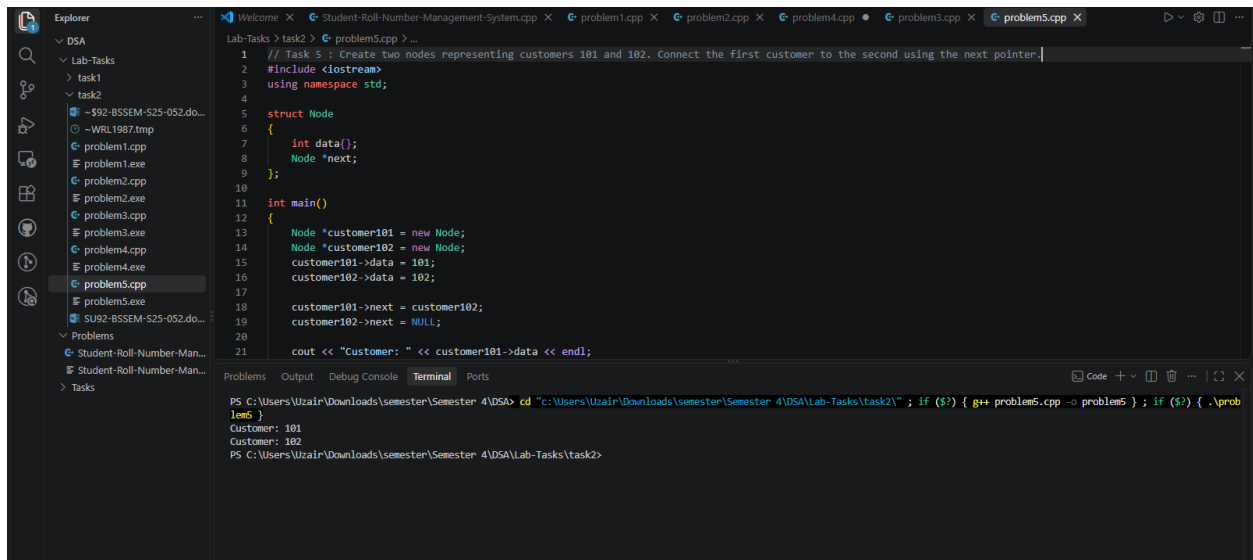
```
cout << "Customer: " << customer102->data << endl;
```

```
delete customer101;
```

```
delete customer102;
```

```
return 0;
```

```
}
```



```
1 // Task 5 : Create two nodes representing customers 101 and 102. Connect the first customer to the second using the next pointer.
2 #include <iostream>
3 using namespace std;
4
5 struct Node
6 {
7     int data[];
8     Node *next;
9 };
10
11 int main()
12 {
13     Node *customer101 = new Node;
14     Node *customer102 = new Node;
15     customer101->data = 101;
16     customer102->data = 102;
17
18     customer101->next = customer102;
19     customer102->next = NULL;
20
21     cout << "Customer: " << customer101->data << endl;
```

```
PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA> cd "c:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks\task2\" ; if ($?) { g++ problem5.cpp -o problem5 }; if ($?) { .\prob
lem5 }
Customer: 101
Customer: 102
PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks\task2>
```